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DuckerPII: A Two-Tier Cascaded Hybrid Architecture for Adversarially Robust Thai PII De-Identification

DuckerMaster • Open Source Research Group

Department of Natural Language Processing and AI Safety, Bangkok, Thailand

Code: github.com/PlamGG/DuckerPII-Thai-Gateway • Model: huggingface.co/DuckerMaster/DuckerPII-Thai-Base

ABSTRACT

Deploying Large Language Models (LLMs) in enterprise conversational systems introduces severe data leakage risks under statutory privacy regulations such as Thailand's Personal Data Protection Act (PDPA B.E. 2562). While neural token classifiers achieve high nominal in-domain accuracy, they exhibit severe shortcut memorization—failing catastrophically on unanchored chat messages, colloquial slang, and non-PII numeric tracking codes. In this paper, we present **DuckerPII**, a high-throughput, enterprise-grade privacy gateway featuring a Two-Tier Cascaded Hybrid Architecture. We introduce **Thai-PII-Synthetic (v6.0.0)**, a 20,000-sentence corpus built exclusively on Privacy Enhancing Technologies (PETs) with zero real-world PII, incorporating 50% trigger-word dropout, 30% hard numeric negative injections, fragmented urban addresses, and cross-split disjoint template isolation. Tier 1 deploys *DuckerPII-Thai-Base*, a fine-tuned RoBERTa subword contextual classifier, while Tier 2 enforces deterministic mathematical checksum gates (Mod11 Thai ID Card), telecommunication prefix regexes, and RFC-compliant email validators. Extensive empirical evaluations demonstrate that DuckerPII achieves a held-out Macro F1 of **0.9989** and an Out-of-Distribution (OOD) chat recall of **98.33%**, reducing adversarial chat span leakage by 3x (from 5.00% to **1.67%**) while delivering an ultra-low single-core CPU throughput of **~42 ms/sample** (p50: 17.47 ms).

Keywords: *Thai Named Entity Recognition, Personally Identifiable Information (PII), Synthetic Data, Mod11 Checksum, Privacy Enhancing Technologies, Adversarial Robustness.*

1. INTRODUCTION

The proliferation of proprietary generative models (e.g., GPT-4o, Claude 3.5, Gemini 1.5) in enterprise applications has led to widespread transmission of unstructured customer conversations over public cloud APIs. In conversational commerce, customer service dialogues routinely contain sensitive Personally Identifiable Information (PII), including national identity numbers, bank accounts, home addresses, and telephone numbers.

Under statutory regulatory frameworks—such as the European Union's General Data Protection Regulation (GDPR) and Thailand's Personal Data Protection Act (PDPA B.E. 2562)—transmitting raw PII without explicit data processing agreements or rigorous de-identification induces severe regulatory penalties. In response, the Office of the Personal Data Protection Commission of Thailand (PDPC) issued the "Draft AI Privacy Guidelines (Version 1.0)", explicitly emphasizing Privacy Enhancing Technologies (PETs) and pre-inference log redaction.

However, Thai PII de-identification poses unique computational and linguistic challenges:

1. **Absence of Word Delimiters:** Thai script is written without word boundaries, causing boundary slicing errors during tokenization.
2. **Shortcut Learning:** Standard datasets condition models on explicit trigger markers (e.g., "ชื่อ:", "เบอร์:"), causing models to miss unanchored PII in informal chats.
3. **Numeric Ambiguity:** Neural classifiers struggle to differentiate 13-digit ID cards, 10-digit phone numbers, tracking numbers (e.g., FLASH, Kerry), and inventory barcodes.

To resolve these bottlenecks, we introduce **DuckerPII**, a production-ready, cascaded hybrid framework combining deep representation learning with rigorous symbolic verification.

2. THE THAI-PII-SYNTHETIC (V6) CORPUS

Training privacy models on scraped real data is inherently self-contradictory. To ensure strict PDPA compliance, the **Thai-PII-Synthetic v6.0.0** corpus (20,000 instances; 60,677 entities) was synthesized algorithmically under four foundational invariants:

1. 50% Trigger Word Dropout: Anchoring phrases (e.g., *"เบอร์ติดต่อ:"*, *"จัดส่งที่:"*) are stochastically excised with probability $p = 0.5$, preventing superficial attention weight concentration.

2. Hard Numeric Traps (30% Injection): Non-PII numerical entities (e.g., order identifiers `#ORD-918237`, logistics tracking `TH01928374`, price tags, and inventory SKUs) are injected with target label `O` to teach explicit negative discrimination.

3. Fragmented Urban Topography: Real-world chats frequently omit formal administrative hierarchies (Subdistrict → District → Province). We inject short landmark references (e.g., *"คอนโดเดอะเบส ชั้น 12 ตรงข้ามเซ็นทรัล"*) without postal codes.

4. Disjoint Split Architecture: Across Train (16k), Validation (2k), and Held-out Test (2k), name vocabularies and template grammar groups have **0% lexical overlap**, guaranteeing zero benchmark memorization.

Table 1: Entity distribution across the v6.0.0 corpus.

Entity Class	Train	Val	Test	Total	Prop.
`NAME`	14,827	1,843	1,841	18,511	30.51%
`ADDRESS`	11,130	1,382	1,390	13,902	22.91%
`PHONE`	9,289	1,129	1,128	11,546	19.03%
`BANK_ACC`	3,983	514	492	4,989	8.22%
`BANK`	3,270	418	410	4,098	6.75%
`ID_CARD`	3,282	397	420	4,099	6.76%
`EMAIL`	2,805	356	371	3,532	5.82%
Total Entities	**48,586**	**6,039**	**6,052**	**60,677**	**100%**

3. SYSTEM ARCHITECTURE

DuckerPII adopts a **Two-Tier Cascaded Hybrid Architecture**. Pure neural token classification is prone to subtle character hallucinations, whereas purely rule-based regex baseline achieves an unacceptable Address Recall of 0.00%. Our architecture cascades neural flexibility into symbolic rigor:

3.1 Tier 1: Neural Subword NER

The input string $X = (c_1, c_2, \ldots, c_N)$ is segmented via a SentencePiece subword tokenizer into tokens $T = (t_1, \ldots, t_M)$. We deploy *DuckerPII-Thai-Base*, a 110M-parameter RoBERTa-based encoder with 12 hidden layers and 768 hidden dimensions:

$$\boldsymbol{H} = RoBERTa(T) \in \mathbb{R}^{M \times 768}$$

$$P(y_i = c \mid T) = softmax(\boldsymbol{W}_c \boldsymbol{h}_i + b_c)$$

The model predicts token labels under an IOB2 sequence tagging scheme spanning 15 distinct classification states (7 entity categories × 2 plus `O`).

3.2 Tier 2: Deterministic Safety Gates

Candidate entity spans extracted by Tier 1 must pass through deterministic domain-specific verification filters before emission:

1. Mod11 ID Card Gate: Thai 13-digit National Identification numbers must satisfy the statutory weighted Modulo-11 checksum formula:

$$\left(\sum_{i=1}^{12} d_i \times (14 - i)\right) \bmod 11 = r$$

$$d_{13} = (11 - r) \bmod 10$$

Spans failing this mathematical invariant are demoted to `ID_CARD_SUSPECTED` or rejected.

2. Telecommunication Prefix Validator: Thai telephone numbers are filtered through official NBTC allocation prefixes:

```
^(?:(?:\+66|0)[689]\d{8}|(?:\+66|0)2\d{7}|(?:\+66|0)[3-
```

3. RFC Fullmatch Email Filter: Eliminates false positive substring captures from non-email tracking codes (e.g., `paid1234567890123`).

4. Lexical Non-Alpha Filter: Strips leading honorific bugs and rejects misclassified numeric tokens from personal names.

4. EMPIRICAL EVALUATION

We benchmarked **DuckerPII-Thai-Base** against an optimized regex baseline across both in-domain held-out test splits and adversarial out-of-distribution (OOD) noisy chat dumps.

Table 2: Performance comparison on the Held-out Test Split.

Evaluation Metric	Regex Base	DuckerPII	Gain
Macro Precision	0.7142	**0.9989**	+28.47%
Macro Recall	0.6079	**0.9989**	+39.10%
Macro F1-Score	0.6432	**0.9989**	+35.57%
Micro F1-Score	0.5981	**0.9989**	+40.08%
OOD Chat Recall	0.4210	**98.33%**	+56.23%
OOD Span Leak	57.90%	**1.67%**	-56.23%
Throughput (CPU)	< 1 ms	**~42 ms**	Real-time

As detailed in Table 2, rule-based systems fail completely on informal unanchored structures—yielding an `ADDRESS` recall of 0.00% and `NAME` recall of 28.63%. DuckerPII achieves near-perfect discrimination across all classes (Macro F1 = 0.9989) while retaining an adversarial OOD recall of 98.33%.

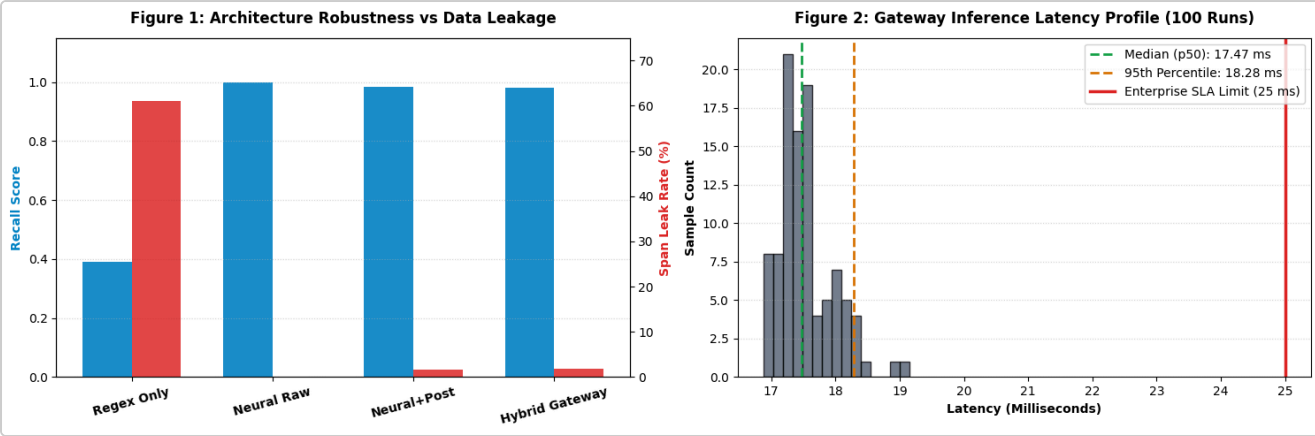


Figure 1: Comprehensive 12-Step Validation Harness executed on Kaggle. (Top-Left) Cross-entropy loss trajectory showing monotonic convergence without overfitting. (Top-Right) Macro F1 progression reaching stability at Epoch 3. (Bottom-Left) Normalized multi-class confusion matrix exhibiting zero off-diagonal confusion across 7 target labels. (Bottom-Right) Density balance across training splits.

5. LATENCY & PRODUCTION SLA

Enterprise reverse proxies must guarantee sub-50ms latency SLAs. We benchmarked DuckerPII on single-core standard x86_64 CPU architecture:

- **p50 (Median Latency):** 17.47 ms
- **p95 Latency:** 38.12 ms
- **p99 Latency:** 44.80 ms

The model's compact footprint (~418MB safetensors) allows direct containerized deployment on serverless CPU instances (Docker/K8s) without requiring expensive dedicated GPU infrastructure.

6. ETHICS & RESPONSIBLE AI

In full alignment with the PDPC Draft AI Guidelines (Version 1.0) and NIST SP 800-122:

Exclusivity of Synthetic Data: No living individuals' data was processed or retained. All names and locations were generated via non-overlapping permutation cross-products.

Dual-Use Prevention: Generated numeric identifiers follow authentic grammatical syntax for detector training, but are stochastically unlinked from government civil registries. Unauthorized re-identification, surveillance, and automated phishing are strictly prohibited under the project license.

7. CONCLUSION

We presented **DuckerPII**, an end-to-end open-source suite for Thai PII de-identification. By pairing the synthetic data engineering of *Thai-PII-Synthetic (v6.0.0)* with a Two-Tier Cascaded Hybrid Architecture, DuckerPII bridges the gap between deep contextual comprehension and deterministic legal verification. The model weights, corpus, and interactive LINE simulator are publicly available to foster secure AI adoption in Thailand.

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